



SIMATS Online Programming Lab

C PROGRAMMING



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CO3 - AT1 - Analytical Problem solving

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QUESTIONS

Q1

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20 marks

**Q2**

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**Q3**

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20 marks

**Q4**

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20 marks

**Q5**

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20 marks



5/5 answered

Q1

CO3 - AT1 - Analytical Problem Solving

20
marks

1. Given an $N \times N$ matrix, Write a C Program to determine whether it is:

- a) Symmetric
- b) Upper triangular
- c) Lower triangular
- d) Diagonal matrix
- e) Identity matrix
- f) Display all applicable classifications.

Your Code

```
1 #include <stdio.h>
2 int main() {
3     int N;
4     scanf("%d", &N);
5     int a[N][N];
6     for (int i = 0; i < N; i++) {
7         for (int j = 0; j < N; j++) {
8             scanf("%d", &a[i][j]);
9         }
10    }
11    int symmetric = 1;
12    int upper = 1;
13    int lower = 1;
14    int diagonal = 1;
15    int identity = 1;
16    for (int i = 0; i < N; i++) {
17        for (int j = 0; j < N; j++) {
```

```
18 if (a[i][j] != a[j][i]) symmetric = 0;
19 if (i > j && a[i][j] != 0) upper = 0;
20 if (i < j && a[i][j] != 0) lower = 0;
21 if (i != j && a[i][j] != 0) diagonal = 0;
22 if (i == j) {
23     if (a[i][j] != 1) identity = 0;
24 }
25     else {
26     if (a[i][j] != 0) identity = 0;
27     }
28     }
29 }
30 printf("Applicable classifications:\n");
31 int any = 0;
32     if (symmetric) { printf("Symmetric\n");
33     if (upper) { printf("Upper triar
34     if (lower) { printf("Lower triar
35     if (diagonal) { printf("Diagonal ma
36     if (identity) { printf("Identity ma
37     if (!any) printf("None\n");
38 return 0;
39 }
```

Input

```
3
1 0 0
0 1 0
0 0 1
```

Output



Run



Save